higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

LOGIC SYSTEMS N6

1 APRIL 2019

This marking guideline consists of 8 pages.

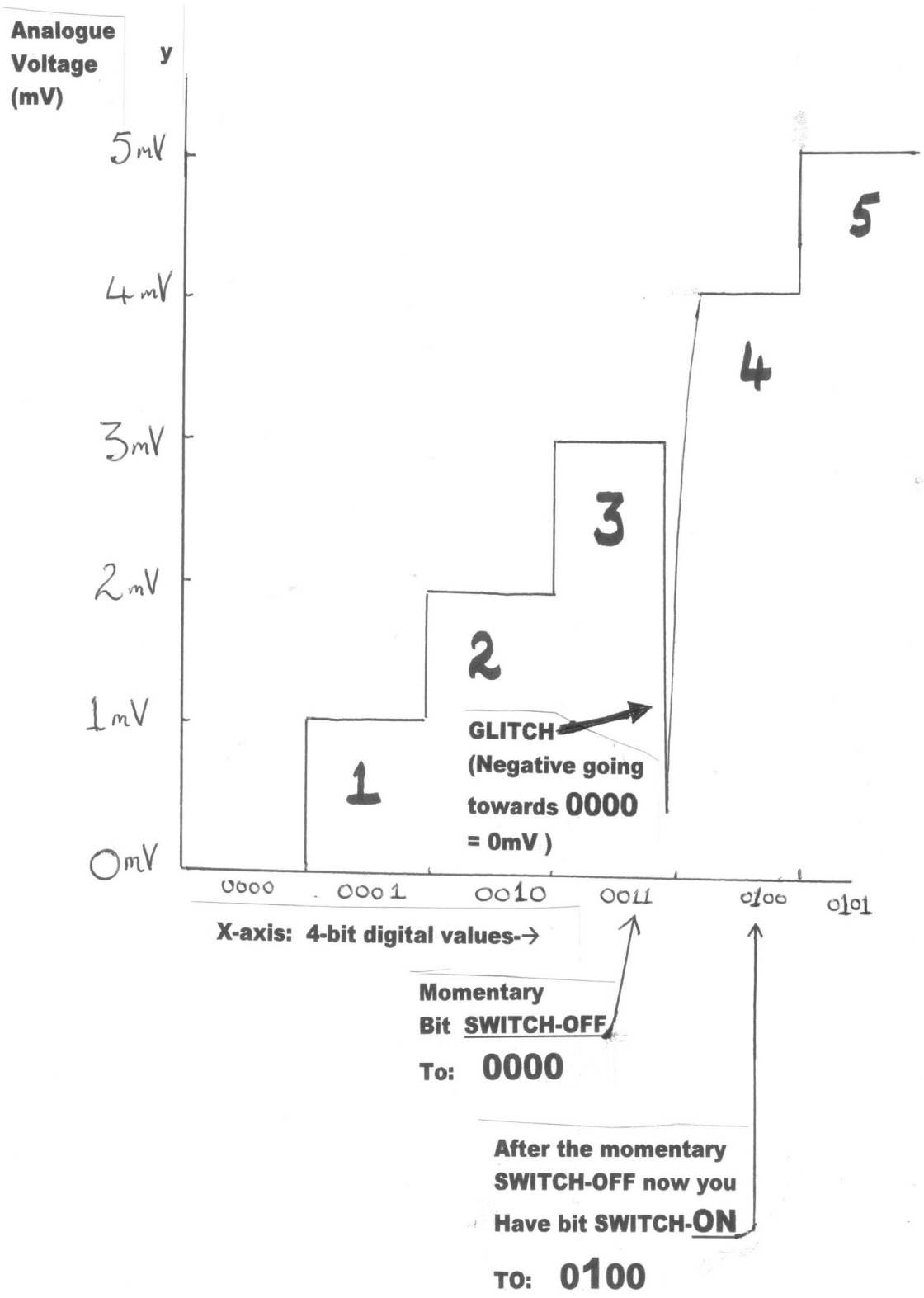
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QUESTION 1

1.1 **H**igh-frequency voltage spikes (sometimes very large) which occur in the output of a digital-to-analogue converter for some changes in the input. (2)

1.2
- Deformation of pulse trains
- Deformation of staircase waveforms
- Disturbance of codes like ASCII (3)

1.3



(8)

1.4 1.4.1 $z = r + r + r + r + v - s - s - s + t + t + t + u$

$z = 4r + v - 3s + 3t + u$

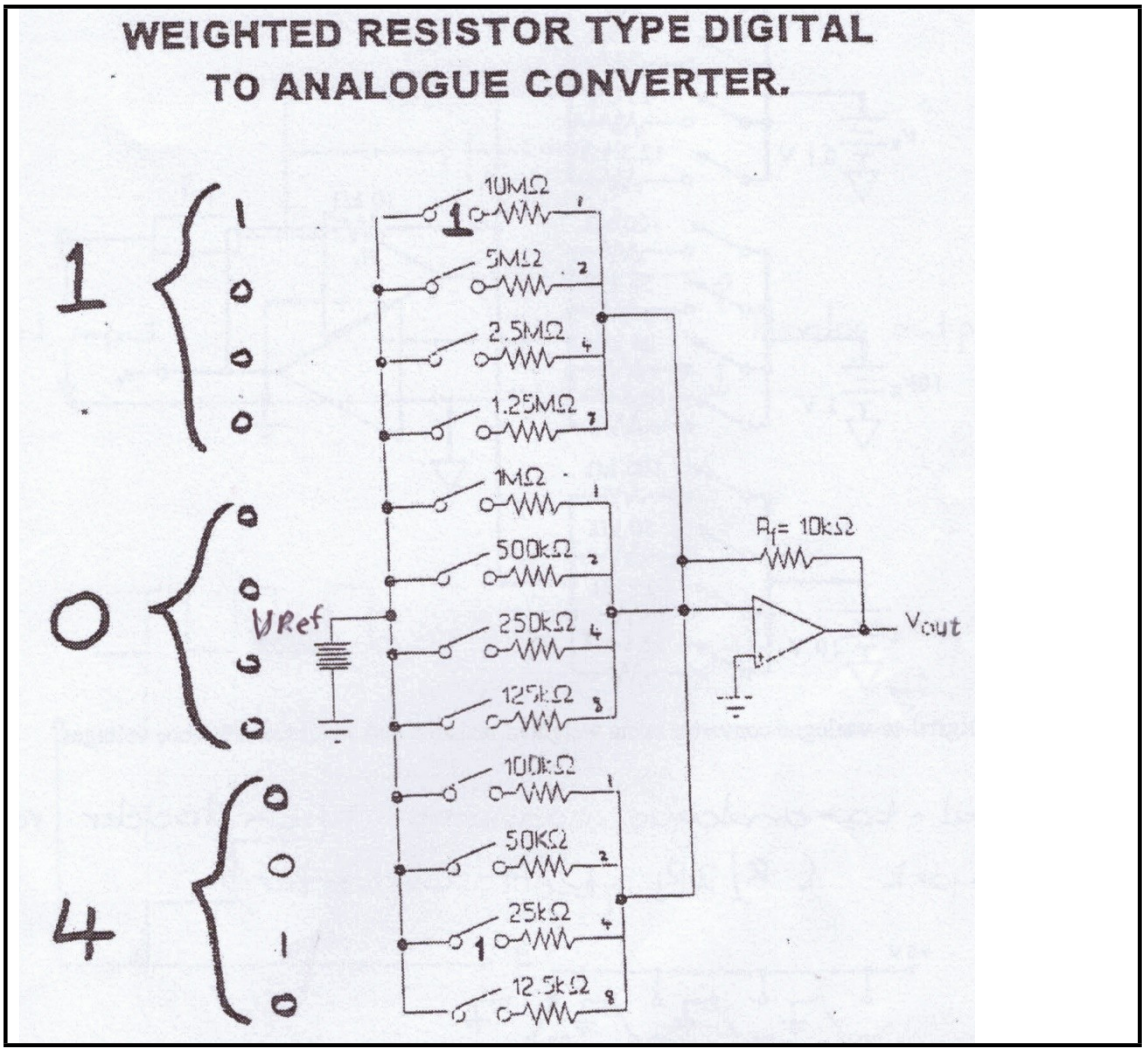
$= 4r + 3t + u + v - 3s$ (5)

1.4.2 ASSEMBLER (which uses mnemonics like: ADDA, MULT (and so ON) (2)

1.4.3 Y = 4(F1B1) + 3(F1B3) + (F1B4) + (F1B5) -3(F1B2)
Y = 4(1100) + 3(800) + (700) + (600) – 3(1000)
Y = 4400 + 2400 + 700 + 600 – 3000
Y = 8100 – 3000
Y = 5100 (DECIMAL) (5)
[25]

QUESTION 2

2.1

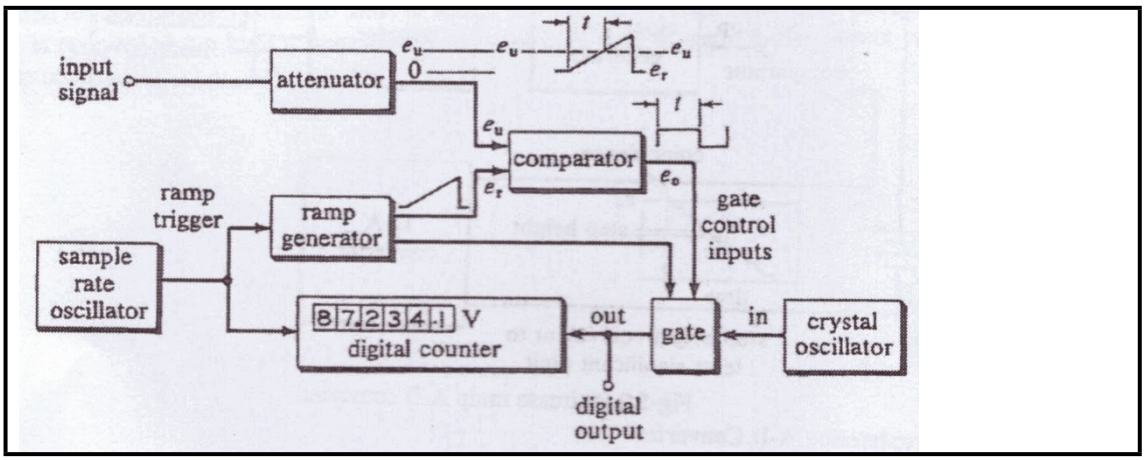


(15)

2.2 (1) LADDER DIAGRAM (GRAPHICAL)
(2) FUNCTION BLOCK DIAGRAM (GRAPHICAL)
(3) STRUCTURED TEXT (TEXT)
(4) INSTRUCTION LIST (TEXT)
(5) SEQUENTIAL FUNCTION CHART (GRAPHICAL) (5 × 2) (10)
[25]

QUESTION 3

3.1 VOLTAGE-TO-TIME-TO-DIGITAL CONVERTER : BLOCK DIAGRAM



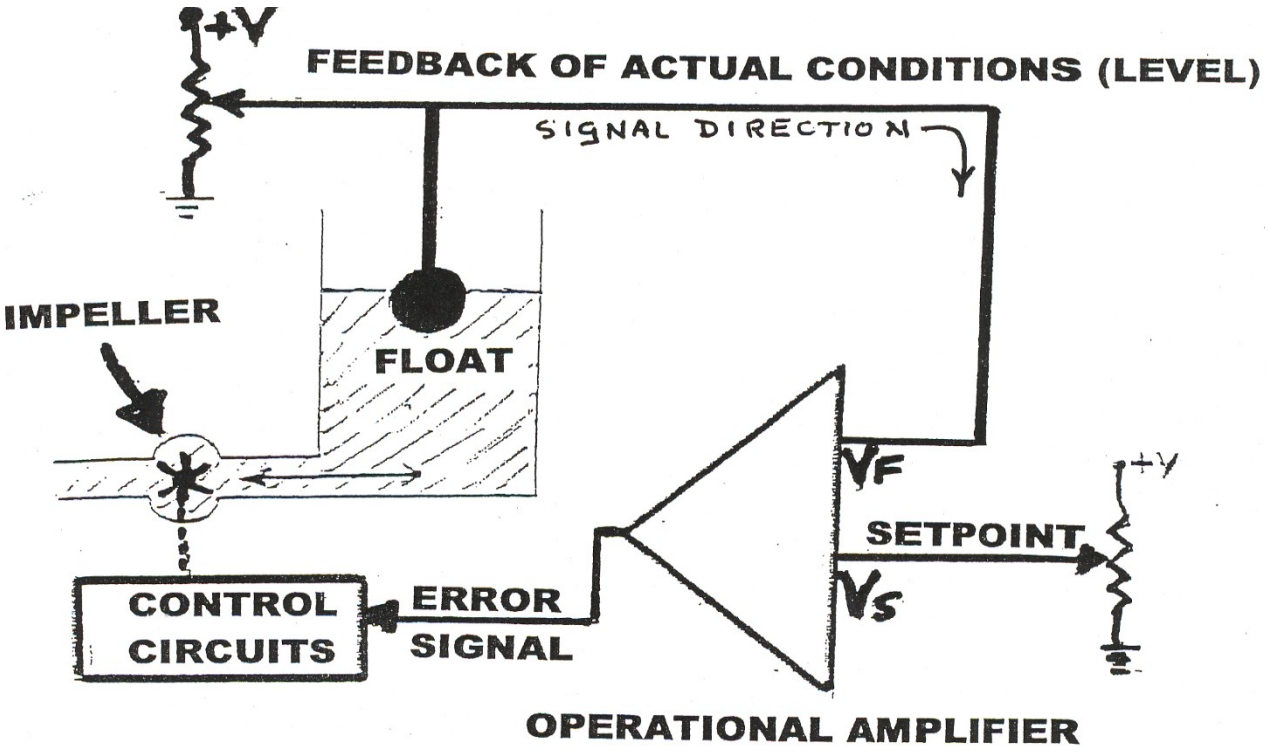
(10)

3.2
- Gate opens when ramp voltage starts its linear sweep
- When ramp voltage equals comparator voltage the comparator
- Closes the gate
- Time interval is now proportional to input voltage
- Time interval clock pulses converted to digital word by counter
- Digital counter displays final count value (5)

[15]

QUESTION 4

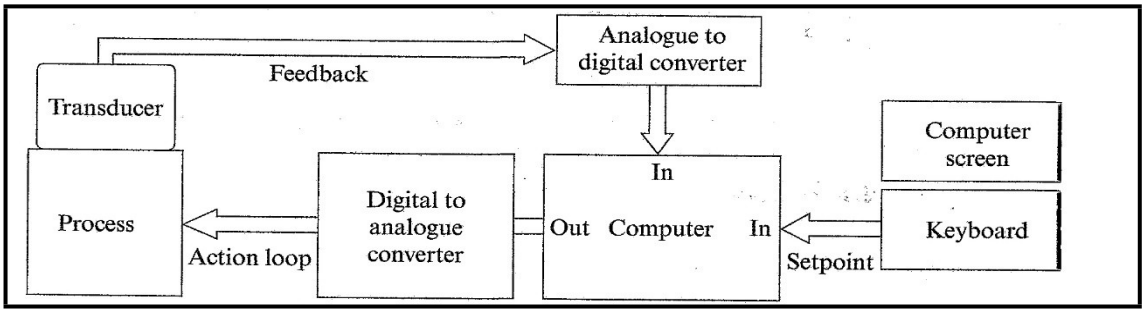
4.1 CONTROL OF TANK-LEVEL USING AN OP-AMP COMPARATOR



The above circuitry and process blocks show an analogue control system using an operational amplifier.

- When VF = S then the ERROR SIGNAL = 0 and the pump impeller does not move.
- When VF ≠ VS than the ERROR SIGNAL causes the bidirectional pump impeller to turn until set-point level is realised. (9)

4.2 COMPUTER INTERFACING



(9)

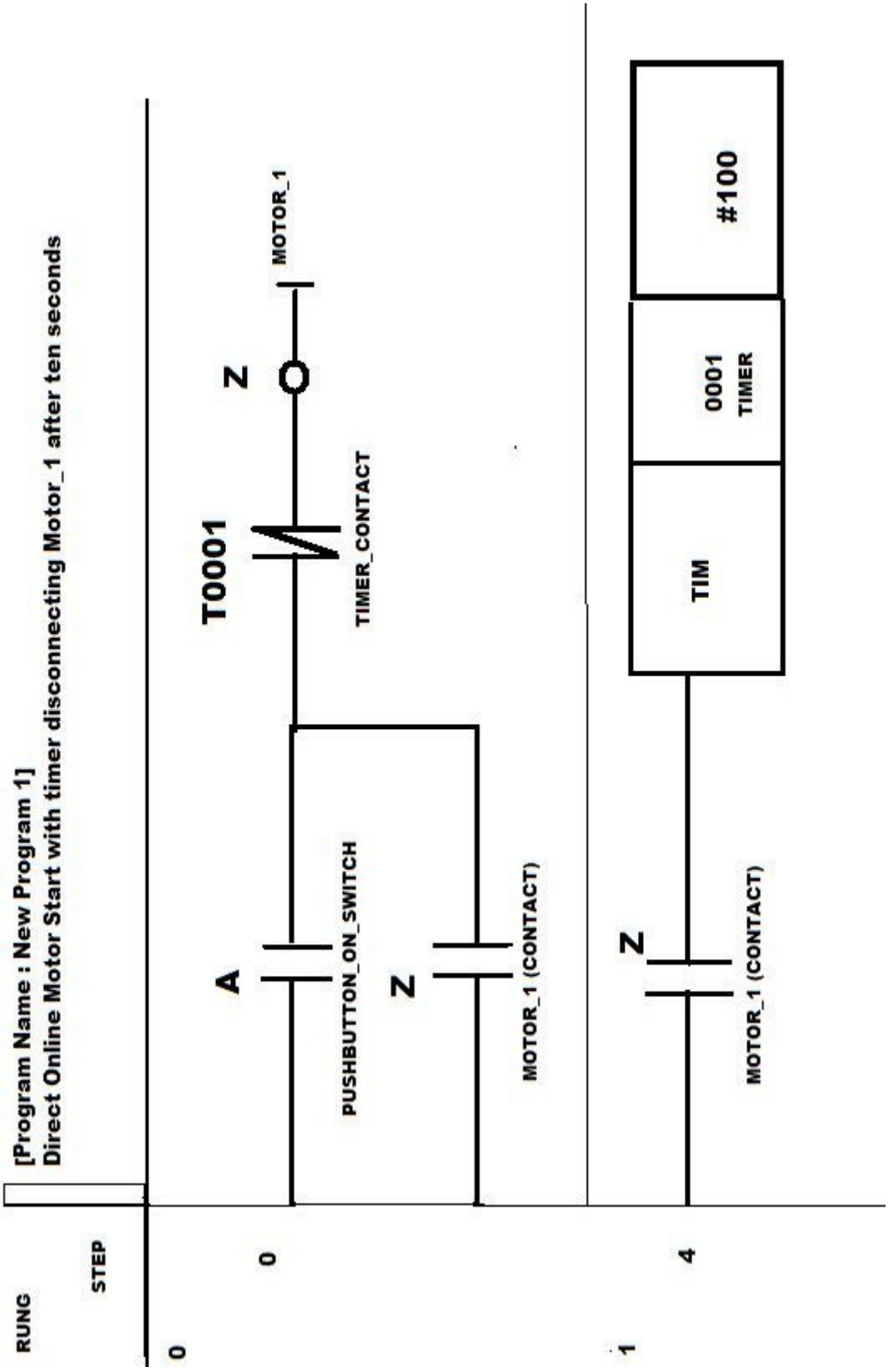
4.3 P.L.C. TOGGLING FROM STATEMENT LIST TO DIAGRAM:

When the statement list mnemonic NOR is displayed in ladder diagram the ladder diagram shows the two **normally closed** inputs in **series.** (2)

[20]

QUESTION 5

5.1 LADDER DIAGRAM



(10)

5.2 The International Electro technical Commission (IEC) is the world's leading organisation that prepares and publishes international standards for all electrical, electronic and related technologies.

Millions of devices that contain electronics and use or produce electricity, rely on IEC International Standards and Conformity Assessment Systems. By complying to IEC International Standards these millions of devices perform, fit and work safely together. South Africans designing or using logic systems (hardware or software) must conform to IEC Standards to run our own industries and to export to the world. The IEC Open International Standard IEC 61131 is of particular importance in this regard. (5)

[15]

TOTAL: 100

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